

Target Variable Transformation in Regression

A Technical Guide on Normalization, Variance Stabilization, and Quantile Scaling

1. Core Motivation & Problem Statement

In regression modeling, predicting continuous numerical targets accurately depends heavily on the distributional characteristics of the target variable (y). Traditional parametric models, such as Ordinary Least Squares (OLS) Linear Regression, implicitly assume that the relationship between features and the target is linear, that the residuals exhibit constant variance (homoscedasticity), and that errors are normally distributed. Real-world datasets (e.g., house sale prices, healthcare costs, or income) frequently violate these assumptions by displaying heavy right-skewness, extreme outliers, and non-linear patterns. Unaddressed target skewness can distort regression coefficients and reduce model generalizability.

2. Primary Benefits of Target Transformation

Benefit Category	Core Mechanism	Impact on Regression Model
Normalization of Distributions	Converts skewed or non-normal target distributions into symmetric, Gaussian-like distributions.	Aligns dataset characteristics with the normality assumption of statistical regression algorithms.
Variance Stabilization	Mitigates heteroscedasticity, where residual spread varies across predictor ranges.	Ensures consistent prediction errors and stable confidence intervals across all values.
Handling Non-Linearity	Maps non-linear predictor-target relationships into linear space using log or power transforms.	Enables linear models to capture complex, non-linear relationships without high model complexity.
Outlier & Performance Control	Compresses extreme tail values into standardized quantile distributions.	Reduces outlier influence, enhancing evaluation metrics like Mean Absolute Error (MAE) and R^2 .

3. Empirical Demonstration: Distribution Shift

Applying non-linear transformations such as **QuantileTransformer** maps raw, right-skewed target distributions (such as Kaggle's *SalePrice*) directly onto a target output distribution (e.g., standard normal distribution). This mapping mitigates the disproportionate influence of high-value properties during model training.

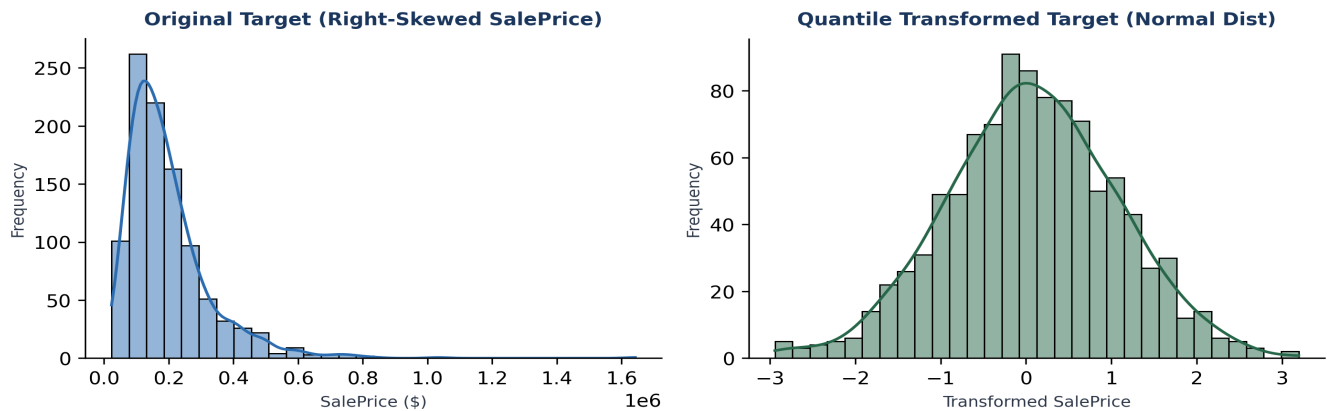


Figure 1: Distribution of raw target vs. QuantileTransformer output (mapped to normal distribution).

4. End-to-End Python Implementation Workflow

The standard workflow in **scikit-learn** involves fitting the target transformer on the training set targets, fitting the estimator on transformed targets, predicting on test features, and finally executing an **inverse transformation** to map predictions back to the original dollar units before computing metric evaluations.

```
# 1. Preprocess Predictors (Imputation & One-Hot Encoding)
X = pd.get_dummies(data.drop('SalePrice', axis=1).fillna(0))
y = data['SalePrice']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 2. Transform Target with QuantileTransformer (Normal Output)
quantile_tf = QuantileTransformer(output_distribution='normal', random_state=42)
y_train_tf = quantile_tf.fit_transform(y_train.values.reshape(-1, 1)).flatten()

# 3. Fit Linear Regression on Transformed Target
model = LinearRegression()
model.fit(X_train, y_train_tf)

# 4. Predict & Perform Inverse Transformation
y_pred_tf = model.predict(X_test)
y_pred_orig_scale = quantile_tf.inverse_transform(y_pred_tf.reshape(-1, 1)).flatten()

# 5. Evaluate Metrics on Original Unit Scale
r2 = r2_score(y_test, y_pred_orig_scale)
mae = mean_absolute_error(y_test, y_pred_orig_scale)
```

5. Best Practices & Critical Pitfalls

CRITICAL RULE — Prevent Data Leakage & Scale Mismatch:

- Strict Data Isolation:** Never call `fit` or `fit_transform` on the full target array or test set. Fit the transformer exclusively on `y_train`.
- Inverse Transformation for Evaluation:** Performance metrics (MAE, RMSE, R^2) must be calculated on predictions transformed back into original units via `inverse_transform()`. Calculating metrics directly on transformed target values yields misleading error figures.